

THEORY ONLY · NO CODE · DAY 1 OF 3

TrustLens

Final Year Project — Panel Viva Preparation

Day 1: The Big Picture

What's in this file: the 60-second pitch to open with, why this project exists (with real numbers), the full data pipeline explained in plain English, and an honest AI-vs-rules status table for all 10 modules — memorize this table before anything else.

Keep this file separate from code testing. This is for reading and re-reading — code changes and live testing belong in a different session with Claude.

Part A — Your 60-Second Opening Pitch

If the panel says “**tell us about your project**” — this is your answer. Say it in your own words, but hit these beats, in this order:

SAY THIS (memorize the beats, not the exact words)

“TrustLens is a web platform that checks whether a social media influencer on Instagram is trustworthy or fake. You give it a username, and in under a minute it gives you a Trust Score from 0 to 100, built from four real checks: is the account itself fake or bot-like, is their engagement genuine or bought, is their content spreading misinformation, and do they falsely claim professional credentials like ‘doctor’ or ‘CFA’. This matters because Pakistan lost billions of rupees to online fraud last year, much of it through influencers making false claims — fake doctors selling supplements, fake financial advisors promising guaranteed returns. We built this to catch that automatically.”

Why this works: it states the *what*, the *how*, and the *why it matters* — in that order — without needing to remember a script word for word.

Part B — Why Does This Project Exist?

The panel will likely ask “**why is this needed?**” You need real numbers, not vague claims.

The problem, in numbers

- Pakistan had **over 13,000 reported online fraud complaints in 2024 alone**.
- Authorities recovered **over Rs. 652 crore** from these cases.
- **False professional identity claims** — someone claiming to be a doctor, financial advisor, or certified expert with no real credential, giving dangerous advice.
- **Influencer ad fraud** — brands paying influencers whose followers or engagement are fake or inflated with bots.

IF ASKED: “Why not just use HypeAuditor or Social Blade?”

Those tools are expensive subscriptions (some \$400+/month), they mostly only detect follower-count spikes, and they don't do deepfake scanning, misinformation detection, or credential checking. TrustLens combines all of that in one free tool.

Part C — The Big Picture: How Data Flows Through the System

Explain this **before** any code, using this factory-line idea: a username goes in one end, a Trust Score comes out the other. Five stations:

1

You type a username

The scan starts from one Instagram username — nothing else is needed.

2

Data Ingestion

We fetch **real** data from Instagram — followers, posts, captions, bio — via the RapidAPI Instagram Scraper. Nothing here is invented.

3

Four analysis modules run, each producing its own score

Fake-Account Detection (AI) · Engagement Analysis (rules) · Misinformation Check (AI + rules) · Credential Extractor (rules).

4

Trust Score Engine

Combines the four module scores using fixed weights (25 / 20 / 25 / 10) into one number from 0 to 100. Weights always add up to exactly 100% — never a placeholder.

5

Results Page

Shows the score, a plain-English explanation of each module, the raw numbers behind it, and lets the user dispute it if they think it's wrong.

ONE SENTENCE TO REMEMBER

“Every score is a weighted average of real, independently-computed signals — nothing is guessed or hardcoded.”

Part D — Module-by-Module Honest Status

This is the **single most important table in this document**. The panel WILL ask “is this real AI or just an API?” for every module. Memorize this before anything else.

#	Module	AI?	Honest one-liner
1	Data Ingestion	No — plumbing	Pulls real data from Instagram via RapidAPI, not invented.
2	Fake-Account Detection	Yes, real AI	Trained Random Forest, 95.8% accuracy on a held-out test set — judges the account's own profile, not its followers.
3	Engagement Analysis	No — rules	Math formula comparing engagement rate to size-based benchmarks. Scope doc only requires “basic NLP” here — satisfied.
4	Misinformation Check	Hybrid: AI + rules	Fine-tuned language model (XLM-RoBERTa), 77% accuracy — but its “scam” label only counts when a hand-written rule confirms real scam wording, because the AI alone over-flagged ordinary ads.
5	Credential Extractor	No — rules	Pattern-matches words like “Dr.”, “MBBS”, “CFA” in the bio. Does NOT verify claims externally (Google billing scare — see Day 2).
6	Trust Score Engine	No — a formula	Weighted average, weights always add up to 100%, no placeholder numbers anywhere.
7	Graph-theory clustering	Yes, real — but offline	The scope doc's exact specified method (Manhattan distance + clustering) — genuinely built, but runs on a research dataset, not live, because Instagram blocks live follower-list scraping.
8	Deepfake Scanner	Does not exist in this repo	Built once, then deliberately deleted. Full honest story in Day 2.
9	Browser Extension	Yes, working	Real Chrome extension using the same AI models as the website.
10	Admin Console	No — a management tool	Handles score disputes, user management, and model monitoring.

■ THE SINGLE MOST DANGEROUS TRAP

Never call Engagement Analysis, Credential Extractor, or the Trust Score formula “AI.” They're legitimate, well-built, and exactly what the scope document asked for — but they're **rules, not machine learning**. If a panelist catches you calling a rule-based module “AI,” it damages your credibility on everything else you say.

Today's Homework (do all three before Day 2)

1

Say the pitch out loud, 5 times

Read Part A out loud five times until it stops feeling memorized — it should sound like you explaining it to a friend, not reciting a script.

2

Recall the AI/not-AI table from memory

Cover Part D and try to say each module's AI-or-not status and one-liner without looking. This WILL be asked in some form.

3

Run one real scan together as a team

Open the Results page, click “Show how this score was calculated,” and get comfortable pointing at real numbers on screen while explaining them out loud.

WHAT'S NEXT

Day 2 goes deeper: why Random Forest over other models, why XLM-RoBERTa over GPT-4, the full honest story of why the deepfake scanner was deleted, how the Trust Score's weight redistribution works, and how to defend the two features that aren't fully live (graph clustering, credential verification).

Day 3 is a mock panel — rapid-fire tough questions with on-the-spot correction.

Prepared for the TrustLens FYP team — Air University Islamabad, Faculty of Computing and AI. Keep this file for re-reading only; code testing happens separately.